

Call Center / Pay-Per-Call Platform System Architecture & Workflow Document

This document provides a structured overview of the proposed Next.js-based Call Center, CRM, Lead Management, Live Call Routing, Billing, Wallet, Analytics, and Admin Platform.

1. High Level Architecture

- Customer

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- Calls Business Number

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- Telnyx / Twilio

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- Incoming Webhook

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- Routing Engine Service

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- Checks Available Agents (State, Zip, License, Wallet, Busy, Priority, Schedule, Skills)

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- Best Matching Agent

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- Ring Softphone

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- Accept / Reject

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- Call Connected

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- Recording + Events + Billing

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- Database + Analytics Dashboard

2. Public Website Flow

- Landing Page

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- Hero

↓

- About

↓

- How It Works

↓

- Why Choose Us

↓

- Testimonials

↓

- FAQ

↓

- Contact

↓

- Footer

3. Lead Generation

- Visitor fills form

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- Phone & Email Validation

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- Duplicate Detection

↓

- Lead Created

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- Lead Stored

↓

- Admin Notification

↓

- Optional Auto Call

4. Agent Registration

- Signup

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- Email Verification

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- Pending Approval

↓

- Admin Review

↓

- Approved

↓

- Dashboard Access

5. Agent Dashboard

- Dashboard

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- Take Calls

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- Call Logs

↓

- Leads

↓

- Wallet

↓

- Referral

↓

- Scripts

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- Tutorials

↓

- Settings

6. Admin Dashboard

- Overview

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- Revenue

↓

- Calls

↓

- Agents

↓

- Charts

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- Top Performers

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- Support

↓

- CMS

↓

- Reports

7. Lead Management

- Lead Created

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- Search

↓

- Filters

↓

- Assign

↓

- Export

↓

- Delete

↓

- History

8. Live Call Routing (Core Engine)

- Customer Calls

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- Webhook Received

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- Routing Engine

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- Availability Check

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- Wallet Check

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- State Match

↓

- ZIP Match

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- Business Hours

↓

- Busy Detection

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- Priority / Round Robin / RTB

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- Best Agent Selected

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- Ring Agent

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- Accept

↓

- Connect Call

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- Recording Starts

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- Duration Tracking

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- Billing

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- Call Ends

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- Save Logs

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- Generate Invoice

9. Wallet Flow

- Recharge Wallet

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- Payment Success

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- Wallet Updated

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- Call Connected

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- Auto Deduction

↓

- Transaction Saved

↓

- Invoice Generated

10. Recording Management

- Call Ends

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- Recording Saved

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- AWS S3

↓

- Recording URL

↓

- Play

↓

- Download

↓

- Delete

11. Billing

- Call Completed

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- Calculate Cost

↓

- Wallet Deduction

↓

- Invoice

↓

- Payment History

12. CRM

- Lead Timeline

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- Tags

↓

- Notes

↓

- Assigned Agent

↓

- Follow Up

↓

- Documents

13. Reporting

- Revenue

↓

- Agent Earnings

↓

- Calls

↓

- Conversion Rate

↓

- Average Call Duration

↓

- CSV

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- Excel

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- PDF

14. User Roles

- Super Admin

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- Admin

↓

- Manager

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- Finance

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- Agent

15. CMS

- Homepage

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- FAQ

↓

- Testimonials

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- Footer

↓

- SEO

↓

- Privacy

↓

- Terms

16. Database Modules

- Users
- Roles
- Permissions
- Agents
- Leads
- Calls
- CallEvents
- CallRecordings
- Wallets
- WalletTransactions
- Invoices
- Payments
- Notifications
- SupportTickets
- Reports
- Settings
- States
- ZipCodes
- Scripts
- Testimonials
- FAQs
- CMSPages
- AuditLogs
- Webhooks
- ActivityLogs

17. Dashboard Modules

Admin

- Dashboard
- Agent Management
- Lead Management

- Call Management
- Wallet
- Revenue
- Reports
- Settings
- CMS
- Support
- Analytics

Agent

- Dashboard
- Take Calls
- Call History
- Wallet
- Scripts
- Referral
- Profile
- Settings

18. API Groups

- Authentication
- Users
- Agents
- Leads
- Calls
- Routing
- Wallet
- Payments
- Reports
- CMS
- Notifications
- Settings
- Analytics
- Webhooks
- Support

19. Recommended Tech Stack

Layer	Recommendation
Frontend	Next.js 16 + TypeScript + Tailwind + shadcn/ui
Backend	NestJS (recommended) or Next.js APIs

Database	PostgreSQL + Prisma
Realtime	Socket.io
Voice	Telnyx / Twilio / Retreaver
Storage	AWS S3
Payments	Stripe
Deployment	Vercel + VPS/Docker

20. Development Roadmap

- Phase 1 - Landing Website & Authentication
- Phase 2 - Agent/Admin Dashboards
- Phase 3 - Lead Management
- Phase 4 - Live Call Routing Engine
- Phase 5 - Wallet & Billing
- Phase 6 - Reports & Analytics
- Phase 7 - CMS & Settings
- Phase 8 - Testing, Security & Deployment

21. Additional Business Modules (Updated)

Sub Admin Panel

- Manage Assigned Agents
- View Reports
- Manage Leads
- Wallet Overview
- Call Monitoring
- Limited Settings Access

Agency Panel

- Agency Dashboard
- Manage Agents
- Leads
- Performance
- Billing
- Payouts
- Reports
- Settings
- Support

Membership Packages

- Basic
- Pro

- Premium
- Enterprise
- Monthly Billing
- Annual Billing
- Feature Limits
- Pricing Control
- Agent Limits

Affiliate System

- Affiliate Registration
- Referral Links
- Referral Tracking
- Commission Management
- Payout Requests
- Marketing Assets
- Performance Reports

Permission Matrix

- Super Admin
- Admin
- Sub Admin
- Agency
- Manager
- Finance
- Agent
- Granular Permissions

22. Updated User Roles

- Super Admin
- Admin
- Sub Admin
- Agency
- Manager
- Finance
- Agent